

Quiz:-

(Linear algebra)

Q: Find the eigen values and eigen vectors.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Step:01 $|A - \lambda I| = 0$

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Step:02

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} = 0$$

Step:03

$$\lambda^3 - \left[\begin{array}{c} \text{Sum of diagonal} \\ \text{element} \end{array} \right] \lambda^2 + \left[\begin{array}{c} \text{Sum of diagonal} \\ \text{minors} \end{array} \right]$$

$$\lambda - |A| = 0$$

$$\lambda^3 - 12\lambda^2 + 40\lambda - 32 = 0$$

$$\lambda = 8, 2, 2 \text{ (eigen values)}$$

Now we find eigen vectors:

For $\lambda = 8$.

$$[A - \lambda I]x = 0$$

$$\begin{bmatrix} 6-d & -2 & 2 \\ -2 & 3-d & -1 \\ 2 & -1 & 3-d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case: 01 put the value of $d=8$.

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-2x_1 - 2x_2 + 2x_3 = 0 \quad \text{--- (1)}$$

$$-2x_1 - 5x_2 - 1x_3 = 0 \quad \text{--- (2)}$$

$$2x_1 - 1x_2 - 5x_3 = 0 \quad \text{--- (3)}$$

Use Cramer's rule eg (1) and (3)

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$2x_1 - x_2 - 5x_3 = 0$$

$$x_1 = -x_2 = x_3$$

$$\begin{vmatrix} -2 & 2 \\ -1 & -5 \end{vmatrix} \quad \begin{vmatrix} -2 & 2 \\ 2 & -5 \end{vmatrix} \quad \begin{vmatrix} -2 & -2 \\ 2 & -1 \end{vmatrix}$$

$$10 + 2 = 12 \quad 10 - 4 = 6 \quad 2 + 4 = 6$$

$$x_1 = x_2$$

$$x_2 = -x_1$$

$$x_3 = x_1$$

$$\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} x_1$$

Q: Show that set of natural number with multiplication is group or not.

* check for group properties:

- ① Closure (Yes)
- ② Associativity (Yes)
- ③ identity element (Yes) $e=1$
- ④ inverse elements. (no) for 2, $\frac{1}{2}$ is not in N .

Therefore, $(N, \times)$ is not a group.

Q: set of integers $(Z, +)$ is an abelian group
abelian group and $(Z, \times)$ is not a group
 $(Z, +)$ - Abelian group check:

- ① Closure (Yes)
- ② Associativity (Yes)
- ③ identity (0)
- ④ inverse $(-a)$
- ⑤ Commutativity $a+b = b+a$

conclusion: $(Z, +)$ is an abelian group.

$(Z, \times)$ - not a group check:

- Tests closure (yes)
Associativity (yes)
identity (1) yes
inverse

for element $2, \frac{1}{2} \notin \mathbb{Z}$ so no inverse.

conclusion: $(\mathbb{Z}, +)$ is not a group.

Q: Show that rational number is field to

Show $(\mathbb{Q}, +, \cdot)$ is a field:

① $(\mathbb{Q}, +)$ is an abelian group

② $(\mathbb{Q} \setminus \{0\}, \cdot)$ is an abelian group

③ multiplication distributes over addition

$$a(b+c) = ab + ac.$$

conclusion: All properties hold so rational number $(\mathbb{Q})$ field.